Critical sparse random graphs violate a forest axiom

Disproof of Conjecture 00000005029

gaochengzhecpu

Abstract

For every n divisible by three, the genuine random graph $G(n, 1/n)$ contains a triangle with probability at least $1/50$. Thus a single first-order forest axiom, the absence of triangles, has probability at most $49/50$ on an unbounded sequence of sizes. It cannot belong to the asserted almost-sure forest limiting theory. The proof uses actual independent edge probabilities and an elementary ordered-pair Bonferroni bound; no Poisson limit or random-graph asymptotic theorem is needed.

The sentence that must hold for a forest theory

In the language of simple graphs, consider

$$\varphi = \forall x\, \forall y\, \forall z\, \neg\big(E(x,y) \wedge E(y,z) \wedge E(z,x)\big).$$

The three adjacencies automatically force the vertices to be distinct, since the graph has no loops. Every forest satisfies this first-order sentence. If the forest axioms formed the limiting theory in the asserted sense, the probability of φ would tend to one.

We bound this single sentence's probability directly. This avoids exchanging the limit with an infinite conjunction of all cycle exclusions. A bound for the actual forest event will also follow, but it is not needed for the sentence-level contradiction.

The actual independent-edge model

Let $h \geq 1$, put $n = 3h$, and use vertices $\{0,1,2\} \times \{0,\dots,h-1\}$. Each unordered pair of distinct vertices is independently present with probability $p = 1/n$. This is exactly $G(n, 1/n)$, the source's $c = 1$ model. No conditioning is imposed.

For an explicit finite construction, choose for every unordered pair an independent uniform label in $\{0,\dots,n-1\}$ and retain the edge precisely when the label is zero. An arbitrary finite prescribed pattern of present and absent edges then has probability $p^a(1-p)^b$, where a, b count its present and absent entries. The Lean project proves this complete pattern formula for the actual mapped graph PMF.

Triangles in three vertex blocks

For each triple $t = (a, b, c)$ with one vertex in each of the three blocks, let E_t mean that its three actual edges are present. There are h^3 triples. The edges are distinct, so independence gives

$$\Pr(E_t) = p^3, \qquad \sum_t \Pr(E_t) = h^3(3h)^{-3} = \frac{1}{27}.$$

Two distinct selected triangles can share at most one edge. If they share an edge, their union has five distinct edges and their joint probability is p^5. Otherwise the union has six edges and the joint probability is p^6, even when a vertex is shared.

There are at most h^6 ordered pairs of triangles of the latter kind. For shared edges, overcount separately by each of the three block-pair types. Sharing an edge fixes the same two corresponding coordinates in the two triples. Choosing those two coordinates and the two remaining coordinates leaves h^4 possibilities. Thus there are at most $3h^4$ ordered pairs sharing an edge. Harmless overcounts, including identical triples in this counting bound, only increase the upper estimate. It follows that

$$\sum_{t \neq s} \Pr(E_t \cap E_s) \leq h^6 p^6 + 3h^4 p^5 = \frac{1}{729} + \frac{1}{81h}.$$

All sums here count ordered pairs, not unordered pairs.

An ordered-pair Bonferroni lower bound

For any finite family of events, the deliberately weakened second Bonferroni inequality is

$$\Pr\Big(\bigcup_t E_t\Big) \geq \sum_t \Pr(E_t) - \sum_{t \neq s} \Pr(E_t \cap E_s).$$

To prove it pointwise, let k be the number of events that occur. If $k = 0$, all terms are zero. If $k \geq 1$, then $k \leq 1 + k(k-1)$, so the sum of indicators is at most the union indicator plus the ordered distinct-pair indicator sum. Taking expectations gives the inequality. This intentionally subtracts both orientations; no factor $1/2$ is missing from the estimate being used.

Combining the two bounds and $h \geq 1$ gives

$$\Pr\Big(\bigcup_t E_t\Big) \geq \frac{1}{27} - \frac{1}{729} - \frac{1}{81h} \geq \frac{17}{729} > \frac{1}{50}.$$

Every selected event witnesses a genuine triangle in the actual graph, so the same lower bound applies to the event that the graph has any triangle.

The asymptotic contradiction

For every $h \geq 1$, therefore,

$$\Pr\big(G(3h, 1/(3h)) \models \varphi\big) \leq \frac{49}{50}.$$

The sizes $3h$ tend to infinity. If the full-size sentence probability tended to one, its restriction to this unbounded subsequence would also tend to one, contradicting the fixed upper bound.

Theorem 1. *At critical sparsity $c = 1$, the forest axioms cannot be the asserted almost-sure limiting theory of $G(n, c/n)$.*

Proof. The fixed first-order forest sentence φ has probability at most $49/50$ along every size $n = 3h$. It therefore fails the necessary probability-one limit for membership in that limiting theory. □

There is also a direct bound for the actual forest event. Three pairwise adjacent vertices produce a genuine closed walk of length three whose intermediate vertices are distinct. It is a cycle, so the graph is not acyclic. Consequently

$$\Pr\big(G(3h, 1/(3h)) \text{ is a forest}\big) \leq \Pr\big(G(3h, 1/(3h)) \models \varphi\big) \leq \frac{49}{50}.$$

The Lean project proves this bridge using an actual Mathlib cyclic walk and its actual acyclicity predicate. It separately proves that neither probability tends to one and that the vertex cardinality tends to infinity.

The source is a conjunction. The false critical forest-theory assertion suffices to refute it; the separate supercritical giant-component and quantifier-depth assertions are outside this submission. No claim that the full graph sequence lacks a probabilistic first-order convergence law is made.

Independent exact checks

The accompanying `verify.py` enumerates all eight simple graphs on three vertices and all 32,768 simple graphs on six vertices, weighting every edge subset by its exact $G(n, 1/n)$ mass. It independently checks the triangle-event bound. For block sizes 1 through 8, it also enumerates all ordered selected-triangle pairs and verifies their five/six-edge union cases and exact overlap counts, using only integer and rational arithmetic. These are supplementary consistency checks; Lean proves the result for every positive block size.

Formalization and reproducibility

The formal vertex set is $\mathrm{Fin}(3) \times \mathrm{Fin}(m+1)$, with its proved cardinality $3(m+1)$. Uniform assignments of labels to unordered pairs are mapped to genuine Mathlib simple graphs. Auxiliary diagonal coordinates do not affect this map: it explicitly forbids loops. Actual event probabilities are taken from the graph PMF, with a proved equality to normalized counting on its original sample space.

The complete present/absent finite-edge pattern law is derived by counting allowed coordinate values. Each present coordinate has one allowed label; each absent coordinate has $n-1$. Normalizing their product proves the independent Bernoulli law on every actual nonloop edge family, rather than only on selected three-edge marginals.

The selected triangle event uses its actual three unordered edges, proved distinct. Unequal triangle indices have union cardinality five or six by symbolic coordinate cases. The overlap counts are actual finite double sums, and the ordered-pair Bonferroni inequality is proved from genuine event indicators. These yield the actual graph triangle lower bound, the literal three-adjacency sentence bound and the forest bound.

Lean formalizes the graph property that is exactly the semantics of the displayed sentence, not separate first-order syntax. The mathematical sentence-level argument is given above; it uses only that single necessary forest axiom, not an interchange of an infinite conjunction with probability limits.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Fresh-build logs, source and hashes are included. Printed dependencies contain only standard foundational axioms. Compile with `tectonic main.tex`. The author agent and parent reviewed the work internally; no external independent review is claimed. The parent developed the exact finite verifier separately.

References

[1] The Last Math Competition, Conjecture 00000005029. Exact bilingual source included as `SOURCE.md`. The finite probability bound and its asymptotic implication are proved directly above and in the accompanying Lean project.